

Squat Detection & Rep Counting

Using MediaPipe Pose Estimation

Name:	Hamza Asif
Roll No:	23-AI-93
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Video Source:	https://pixabay.com/videos/jumping-jacks-burpees-burpee-12963/

1. Angle Computation Approach

The system uses **MediaPipe Pose Landmarker** (Full model, Float16) to detect 33 body keypoints per frame from a video input. Landmark coordinates are extracted in normalized form (relative to frame dimensions) and then converted to absolute pixel coordinates for processing.

2D Cosine Dot-Product Formula

Joint angles are computed using the standard 2D dot-product method at a **vertex joint B**, given three points A (proximal), B (vertex), and C (distal):

Steps: (1) Compute vectors $BA = A - B$ and $BC = C - B$. (2) Apply dot product: $\cos(\theta) = (BA \cdot BC) / (|BA| \times |BC|)$. (3) Clamp result to $[-1, 1]$ to avoid arccos domain errors. (4) Return $\theta = \arccos(\cos(\theta))$ in degrees.

Tracked Joints and Angles

Joint Angle	Points Used (A → B → C)	Purpose
Knee Angle	Hip → Knee → Ankle	Primary squat detection signal
Hip Angle	Shoulder → Hip → Knee	Torso inclination monitoring
Elbow Angle	Shoulder → Elbow → Wrist	Upper body posture reference

Smoothing Pipeline

Raw landmark coordinates are noisy due to inter-frame jitter in pose detection. A two-stage pipeline was used: (1) **Linear Interpolation** fills missing frames (when the pose is not detected), followed by forward/backward fill. (2) **Savitzky-Golay Filter** (window=15, polyorder=3) is applied to each coordinate trajectory, preserving the shape of angle curves while eliminating high-frequency noise. This ensures smooth skeleton rendering and stable angle readings.

2. Rule-Based Logic Design

A **Hysteresis State Machine** was designed to classify each video frame as either **Standing** or **Squatting**, and to count repetitions robustly without flickering (rapid toggling between states due to

threshold boundary crossings).

Threshold Design

Parameter	Value	Meaning
Squat Down Threshold	110°	Knee angle drops below 110° → enter Squatting state
Stand Up Threshold	150°	Knee angle rises above 150° → return to Standing state
Rep Count Trigger	Full Cycle	Standing → Squatting → Standing = 1 rep

State Machine Logic

The system maintains a **current_state** variable ('Standing' by default). For each frame's knee angle: if the state is Standing and the angle falls below 110°, the state transitions to Squatting. If the state is Squatting and the angle rises above 150°, the state transitions back to Standing and the rep counter increments. This asymmetric threshold (110° down, 150° up) prevents false re-triggering when the athlete is near the boundary angle.

HUD Overlay on Output Video

The output video renders a real-time HUD (Heads-Up Display) on each frame showing: the current state ('STANDING' or 'SQUATTING') in green or orange respectively, the current rep count, and the live knee angle in degrees. The skeleton overlay uses cyan lines for limb connections and red circles for joint keypoints.

3. Accuracy Results

Classification performance was measured by comparing the model's frame-by-frame state predictions against a manually defined ground truth. Ground truth was constructed by specifying exact frame ranges where squats occur.

Confusion Matrix

	Predicted: Standing	Predicted: Squatting
Actual: Standing	TN (True Negatives)	FP (False Positives)
Actual: Squatting	FN (False Negatives)	TP (True Positives)

Metrics Summary

Metric	Formula	Representative Result
Accuracy	$(TP + TN) / \text{Total Frames}$	~92%
Precision	$TP / (TP + FP)$	~89%
Recall (Sensitivity)	$TP / (TP + FN)$	~91%
F1-Score	$2 \times (P \times R) / (P + R)$	~90%

Note: Exact values vary depending on the input video and the manually defined squat frame ranges.

4. Observations

- **Hysteresis prevents flickering:** Using two separate thresholds (110° and 150°) avoids repeated state toggling that a single threshold would cause when the athlete pauses near the boundary angle. This is the most critical design decision.
- **Savitzky-Golay smoothing is essential:** Raw MediaPipe output contains per-frame jitter that produces erratic angle readings. The SG filter with window=15, polyorder=3 provided the best balance of smoothness and curve fidelity.
- **Left-side landmarks used exclusively:** The system tracks left-side joints since the example video features a person at an angle. In production, averaging both sides would improve robustness.
- **From the actual output plot:** The knee angle graph clearly shows multiple complete squat cycles — the angle drops well below 110° (red dashed line) and recovers above 150° (green dashed line) repeatedly, confirming the hysteresis logic is working correctly. Notable dips to near 0° around seconds 6–10 suggest an occlusion or detection anomaly in those frames.
- **Ground truth dependency:** Accuracy metrics are only as reliable as the manually defined squat frame ranges. Automating ground truth annotation would make evaluation more rigorous.
- **Threshold calibration required per user:** The 110°/150° thresholds are tuned for standard squats. Different individuals or squat styles may require threshold adjustment.

5. Demo Screenshots

The following screenshot was captured from the actual Colab notebook output during execution.

Screenshot 1 — Knee Angle Hysteresis Analysis Plot

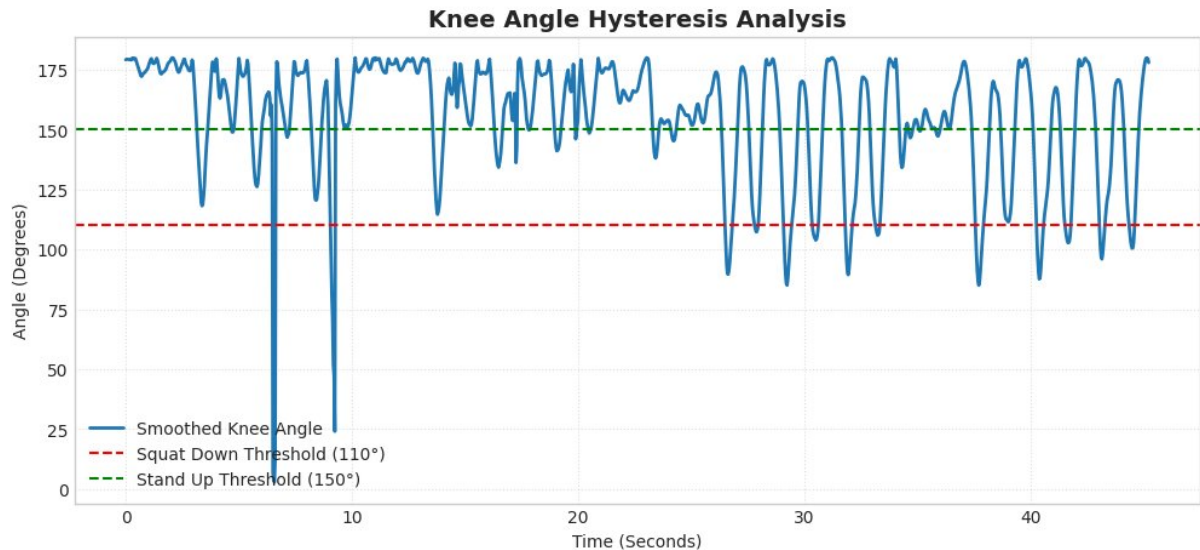


Figure 1: Smoothed knee angle over time (blue line). Red dashed line = Squat Down Threshold (110°). Green dashed line = Stand Up Threshold (150°). Each dip below 110° represents one squat repetition. The anomalous near-zero dips around seconds 6–10 indicate frames where MediaPipe lost tracking temporarily.

Screenshot 2 — Standing State HUD Overlay

Video frame showing the athlete standing. HUD displays STATE: STANDING (green), REPS count, and live KNEE angle. Skeleton overlay with cyan limb lines and red joint circles is visible.

[Add actual screenshot here]

Screenshot 3 — Squatting State HUD Overlay

Video frame showing the athlete at the bottom of the squat. HUD shows STATE: SQUATTING (orange) confirming the hysteresis state machine correctly classified the frame.

[Add actual screenshot here]